



Qualcomm DEVC system being installed underneath the surface of a test track
(Credit: www.qualcomm.com)

German company, has made available commercially a 3kW wireless charging system for industrial applications with an overall transfer efficiency exceeding 90%.

Dynamic charging systems. The crowning glory of the wireless charging technology is its ability to power a car traveling at speeds as high as 110 kilometres per hour as it drives over chargers embedded into the surface of the road.

Qualcomm has developed a system, called Dynamic Electric Vehicle Charging (DEVC), in which a wireless charging system is installed underneath the surface of a 100-metre test track, to provide charging power to a fleet of specially modified electric vans, each fitted with two 10kW charging pads. At a time, several vehicles can be driven along the road, each taking a small amount of energy from each charger as they drive by. Though the batteries do not get fully refilled, they essentially do not lose any power while using the road and the stored energy is saved for an unpowered road. The system is robust as it does not get affected by bad weather and works equally well when the track is covered with water.

Qualcomm is experimenting fitting DEVC under the surface of the racing track of a Formula E circuit. The drivers would have to decide between taking the faster conventional racing line or a longer DEVC fitted route where no stoppage is required for topping up batteries.

When chargers embedded into the roads become a commercial viability,

the impact would be revolutionary and crucial for autonomous cars of the future. Instead of taking the EVs to the charger every few hours, these would be constantly charged from the road as they drive along. When the dynamic wireless charging systems become a commercial viability, it would be a crucial milestone for autonomous cars of the future as these would be constantly charged while driving.

Application areas

Applications of wireless charging can be divided into two broad areas: low power and high power. Low power applications generally include small consumer electronic devices such as cell phones, handheld devices, small computers, and other devices that need charging at power levels below 100 watts. High power wireless charging refers to charging at power levels above one kilowatt.

Power levels of the charging systems that use resonated primary and secondary coils range from 1kW to 300kW. Their most significant application area is in electric EVs, but it is not limited to EVs.

Recently, Japan-based robot manufacturer Daihen has developed a wireless power transfer system for automatic guided robots. Using this wireless charging system, the robots simply need to pull up to a wireless charging area to power up and then go about their duties.

Wireless charging is increasingly finding application in the medical sector as well. The technology enables long-term charging of implants

and sensors located beneath the skin. It has been possible to print wireless power transmitting antennas on flexible materials placed under the skin of patients. These under-skin materials enable monitoring the patient status over a longer period, leading to a more detailed diagnosis by the doctors.

Advantages of wireless charging

Newer trends in wireless charging technology are making available more efficient and compact charging systems. These systems use ultra-thin coils, higher operating frequencies, and optimised drive electronics to reduce the transfer losses. The systems can be easily integrated into the mobile devices or batteries.

The major advantage of wireless charging is that there is no possibility of electric shock, as there are no exposed conductors. Some other advantages are:

- Wireless charging at high power levels enables charging of EVs while in motion, consequently extending the driving range
 - Provides higher reliability as the charging systems are operated automatically without dependence on people to plug, unplug, and operate
 - Less likelihood of electrical faults, such as short circuits
 - Protection against danger of corrosion as circuits and devices are enclosed, providing shield from water and oxygen in the atmosphere
 - Significantly less wear and tear as there is no need to constantly plug and unplug the device, providing greater durability
 - Increased convenience and aesthetic quality as no cables are used
 - Low infection risk for embedded medical devices as transmission of power via a magnetic field passing through the skin precludes the infection risks associated with wires piercing the skin
- However, the inductive wireless charging systems using low power (less than 100 watts) may have the following disadvantages:
- Due to the slower charging and lower efficiency, the devices take longer to charge with the same supplied power
 - Drive electronics and coils